

---

# Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 Quantizing a large language model reduces serving cost and costs some accu-  
2 racy. Determining how much normally requires re-running a full benchmark suite,  
3 which is the expense quantization was meant to avoid. We ask whether the accu-  
4 racy delta can be predicted instead. Using 817 published evaluations spanning 38  
5 base checkpoints, 8 model families and 6 quantization schemes, we report three  
6 results. First, a *negative* one: per-model prediction carries almost no signal. Under  
7 leave-one-family-out validation the best learned predictor beats a global mean by  
8 0.03pp of MAE, and both ridge regression and gradient boosting are *worse* than  
9 guessing the average, because 33–100% of the variance on most benchmarks is  
10 evaluation noise. Second, a positive one: conditioning only on the quantization  
11 scheme yields split-conformal intervals with 90.1% empirical coverage (118/131,  
12 95% CI [83.6, 94.6]) on checkpoints never used in construction. Third, and most  
13 consequential for practice: the published corpus is outcome-truncated. Every con-  
14 figuration in it is one that someone chose to release. We rented an A100 and mea-  
15 sured deliberately-faulted configurations, observing losses to  $-39.5$ pp against a  
16 published worst case of  $-8.86$ pp—measured, we stress, only on models of 1.5B  
17 and below, and on configurations engineered to fail rather than typical ones. We  
18 argue the resulting artifact should be described as a calibrated historical baseline  
19 rather than a predictor, and we state precisely why its coverage guarantee does  
20 not transfer to a practitioner’s own untuned recipe. Code, data and every figure’s  
21 provenance are public.

## 22 1 Introduction

23 Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8  
24 wants one number: how much accuracy will this cost? The honest answer today is that nobody  
25 knows without running the evaluation.

26 This paper asks whether that number can be forecast from the public record. Model publishers  
27 release quantized checkpoints with benchmark tables comparing the compressed model against its  
28 dense parent. Aggregated, these are thousands of paired measurements of exactly the quantity of  
29 interest.

30 Our contributions:

- 31 1. **A negative result** (§5). Per-model prediction of the accuracy delta does not work, and  
32 we show why: the target is dominated by evaluation noise. We report this as the primary  
33 finding rather than burying it.
- 34 2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured  
35 coverage on held-out families and an explicit refusal rule for cells where coverage is poor.

- 36 3. **A measurement of the corpus’s own selection bias** (§7), via GPU experiments on  
37 deliberately-faulted configurations—data the published record structurally cannot contain.
- 38 4. **An audit** (§9) recording every error we found in our own work, including two that changed  
39 published claims.

## 40 2 Related Work

41 **Predicting evaluations.** BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism  
42 closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and  
43 decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks  
44 is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is a  
45 model’s absolute score; ours is the paired delta between a model and its compressed self, and their  
46 corpus contains no quantized checkpoints.

47 **Conformal prediction on compressed models.** Tong et al. Tong et al. [2026] apply conformal  
48 prediction directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The  
49 estimand differs: their conformal sets are over label space, constructed from a compressed model’s  
50 own output probabilities, so the method measures uncertainty *after* running the model. Ours is con-  
51 structed over historical accuracy deltas to give an interval *before* running anything. Their Figure 2  
52 plots the same quantity we predict,  $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$ , but measures it. Their finding that com-  
53 pression decouples accuracy from uncertainty bounds what an accuracy-only estimate like ours can  
54 claim (§8).

55 **Quantization methods.** The checkpoints we harvest are produced with GPTQ Frantar et al. [2022]  
56 and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024].

57 **The source of our data.** Kurtić et al. Kurtić et al. [2024] report over 500,000 evaluations of  
58 quantized Llama-3.1 models, finding recovery above 99% of baseline. That work does not address  
59 whether the published set of quantized models is a biased sample. Characterising that gap is the  
60 contribution of §7.

61 **Selection bias.** Survivorship bias in benchmark corpora has been formalised and quantified in  
62 information retrieval Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to  
63 fail under selection: conditional on a selection event, calibration data are no longer exchangeable  
64 with a test point Barber et al. [2023], Jin and Candès [2024]. We apply this directly to our own  
65 artifact in §8.

## 66 3 Data

67 We harvest published model cards into rows of (model, scheme, benchmark,  $\text{acc}_{\text{before}}$ ,  $\text{acc}_{\text{after}}$ ), yield-  
68 ing 817 rows after filtering, across 38 base checkpoints, 8 families and 6 schemes.

69 **An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair.  
70 This is redundant information, and we exploit it: for every three-column row we check recovery  $\approx$   
71  $100 \times \text{after}/\text{before}$  to a tolerance derived from the card’s own printed precision, and reject rows that  
72 disagree. This caught fabricated values arising from column-order variation between publishers, and  
73 internally inconsistent cards. Rejections are recorded rather than silently dropped.

74 **A structural limitation, stated at the outset.** All rows come from a single publisher using a  
75 single toolchain. Holding out a model family tests transfer across *models*; nothing in this design  
76 tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

## 77 4 Method

78 **Split conformal.** Given calibration residuals  $\{s_i\}_{i=1}^n$  and level  $\alpha$ , the interval half-width is the  
79  $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which gives exact finite-sample coverage under exchangeability.  
80 Where  $n$  is too small for that index to exist, we return  $\infty$  rather than a misleadingly finite band.

81 **Mondrian (group-conditional) calibration.** A single marginal interval advertised at 90% was  
 82 in fact a 68% interval for NVFP4 users—the users who most need it. We therefore calibrate per  
 83 scheme.

84 **A refusal rule.** For each (scheme, size-band) cell we measure coverage directly. Where mea-  
 85 sured coverage falls below a threshold on a sufficient number of scored rows, the tool emits  
 86 INSUFFICIENT CALIBRATION and no interval.

87 The refusal is scored on *one-sided* coverage: how often the true delta stayed at or above the inter-  
 88 val’s lower bound. The risk this tool bounds is accuracy loss, and a model that beats its envelope  
 89 has exposed no one to anything, so counting that as a failure would refuse cells that are merely out-  
 90 performing. Two cells are refused:  $w_{4a16} < 2B$ , where losses stayed above the lower bound 70.1%  
 91 of the time over 77 rows from 2 checkpoints (two-sided containment 68.8%), and  $w_{8a16} > 10B$  at  
 92 81.1% over 217 rows from 5 checkpoints (two-sided 73.7%). The distinction was identified by an  
 93 external reader. Comparing a one-sided measurement against the printed 90% is the *lenient* reading,  
 94 not a flattering one: the interval is symmetric, so the permitted 10% is split across two tails, and  
 95 pooled one-sided coverage across all scored rows is 95.1%. Both cells are refused even under the  
 96 lenient comparison.

97 **A support floor.** Coverage measured on many benchmarks from one checkpoint is not indepen-  
 98 dent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct  
 99 checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

## 100 5 The negative result

101 Under leave-one-family-out validation, row-weighted MAE:

|     | predictor                       | MAE (pp) | beats global mean? |
|-----|---------------------------------|----------|--------------------|
|     | per-scheme mean (shipped)       | 0.7227   | yes                |
|     | global mean (baseline)          | 0.7545   | —                  |
| 102 | per-(scheme $\times$ benchmark) | 0.7607   | <b>no</b>          |
|     | ridge, 38 features              | 0.7639   | <b>no</b>          |
|     | per-benchmark mean              | 0.7647   | <b>no</b>          |
|     | gradient boosting               | 0.7878   | <b>no</b>          |

103 Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size,  
 104 bit-width, method, benchmark identity and headroom all degraded out-of-family performance.

105 **Why.** Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta  
 106 should be approximately zero—gives a noise floor that accounts for 33–100% of observed variance  
 107 on most benchmarks. On GSM8K the noise alone is  $\pm 2pp$ , while the quantity being predicted has  
 108 mean  $-0.36pp$ . The target is a difference of two noisy quantities that largely cancel: the numerically  
 109 smallest and hardest-to-predict summary of quantization damage available.

110 **Implication.** The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band,  
 111 with no model and no conformal machinery, achieves comparable coverage. What conformal con-  
 112 tributes is a finite-sample guarantee and a principled rule for small- $n$  schemes, not accuracy. We  
 113 therefore describe the result as a *calibrated historical baseline*, not a predictor.

## 114 6 Results

|     | scheme      | 90% interval (pp) | worst observed | % losing >3pp | evals |
|-----|-------------|-------------------|----------------|---------------|-------|
|     | fp8         | [-1.11, +0.48]    | -1.90pp        | 0.0%          | 80    |
|     | w8a8_int    | [-1.71, +1.07]    | -3.32pp        | 1.0%          | 194   |
| 115 | fp8_dynamic | [-1.44, +1.34]    | -8.72pp        | 1.0%          | 200   |
|     | w8a16       | [-0.98, +0.81]    | -3.04pp        | 1.2%          | 86    |
|     | w4a16       | [-2.87, +1.40]    | -8.86pp        | 5.7%          | 193   |
|     | nvfp4       | [-4.10, +1.87]    | -8.12pp        | 15.6%         | 64    |

116 **Prospective validation.** After freezing the method we harvested checkpoints not used during de-  
 117 velopment. Under a strict protocol that excludes any family present in training and any card whose  
 118 format informed a parser fix, coverage is  $118/131 = 90.1\%$ , 95% Clopper–Pearson CI [83.6, 94.6],  
 119 against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code  
 120 with the pipeline reproduces this figure with zero disagreements.

121 **What the envelope cannot do.** 0 of 6 intervals exclude zero: the tool can never state that a scheme  
 122 *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when  
 123 real damage occurs, the interval usually misses it. This is the single most important caveat in the  
 124 paper.

## 125 7 The measured left tail

126 The corpus contains only configurations someone chose to publish. To measure what it structurally  
 127 cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

128 **Result.** Worst measured loss  $-39.5\text{pp}$ , against  $-8.86\text{pp}$  as the worst loss anywhere in the pub-  
 129 lished corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-  
 130 configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor  
 131 instead of per-group.

132 **Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes  
 133 that catastrophic damage is reachable; it is not a measured bound for larger models.

134 **A confound we found and did not resolve.** Re-running MMLU under the published protocol  
 135 reproduces published base accuracy (47.20 against 47.42; 59.57 against 60.98), closing the harness  
 136 question. It exposed a larger one: our own *correctly-configured* control is  $2.6\times$  more damaging than  
 137 published W4A16 on closely comparable checkpoints (mean gap  $-3.78\text{pp}$ ,  $z = 2.9$  and  $5.3$ ). The  
 138 cause is approximately  $171\times$  less calibration data than production defaults, not a bug. Consequently,  
 139 excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It is a  
 140 conservative anchor, not a bound, and we decline to publish a corrected point estimate: the correction  
 141 moves by  $5.70\text{pp}$  depending on the anchor set, and we have only 6 rows from two models.

## 142 8 Limitations

143 **The guarantee is conditional, and void where it matters most.** Split conformal gives exact cov-  
 144 erage under exchangeability between calibration and test. Our calibration set is the set of configura-  
 145 tions a publisher chose to release—a selection event. Conditional on selection, calibration points are  
 146 not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al. [2023],  
 147 Jin and Candès [2024]. It holds for a new checkpoint drawn from the same publication process. It is  
 148 **void** for a recipe the reader tuned themselves and that nobody would have published—which is the  
 149 case a practitioner most wants covered.

150 **Accuracy is not the whole of deployment risk.** Tong et al. [2026] show compression  
 151 frequently decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets  
 152 inflate. A scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We  
 153 measure none of that.

154 **One publisher, one toolchain, and why.** See §3. Our own control arm (§7) is direct evidence  
 155 that practitioner and calibration volume can dominate scheme choice. We measured how confining  
 156 this is: surveying 5 publishers and 195 quantized model cards, 5 of 157 cards from the other 4  
 157 publishers carry paired before/after evaluations, so others do publish them. What no other publisher  
 158 in the sample provides is a *Recovery* column, which is what lets each row be checked against  $100 \times$   
 159 after/before and rejected on disagreement: of their rows, 0 pass that gate, against 142 from the  
 160 control publisher. The corpus is single-publisher not because others publish nothing, but because  
 161 only one publishes in a self-verifying form.

162 **Rows are not independent.** Up to sixteen benchmark rows come from a single quantization run.  
 163 We mitigate with group-conditional calibration and the support floor, and note that a cluster bootstrap  
 164 over checkpoints does not materially widen the headline interval, but the unit of analysis remains a  
 165 row.

## 166 9 Audit

167 We record errors found in our own work, including those that changed published claims. This  
 168 section exists because the alternative—reporting only what survived—would misrepresent how the  
 169 result was obtained.

### 170 Errors that changed a published number.

- 171 • *A retracted headline.* An early claim that held-out models came from families never used  
 172 in construction was wrong; they were new checkpoints of families already present. The  
 173 strict figure is 118/131.
- 174 • *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was  
 175 matched as Math-Lvl-5, contaminating 7 test rows with scores from a different bench-  
 176 mark.
- 177 • *A centring mismatch.* Calibration measured residuals about the stratum mean while in-  
 178 tervals centred on the scheme mean, mis-sizing every calibrated width. Found only by  
 179 independent reimplementations.
- 180 • *A column-order fault.* One publisher orders columns [recovery, base, quant], producing a  
 181 fabricated  $-68\text{pp}$  delta. The integrity gate rejected 10 of 11 rows on that card, passing only  
 182 the one where base = quant made the check degenerate.

183 **A regression we nearly shipped.** An audit indicated that a raw asymmetric empirical band cov-  
 184 ered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization  
 185 damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch.  
 186 On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-  
 187 sided empirical index requires more calibration points than are often available. An infinite interval  
 188 covers by construction. On rows where both are finite, the original wins on coverage *and* width. The  
 189 change was reversed.

190 **An overclaimed guarantee.** The artifact was described as carrying a coverage guarantee without  
 191 qualification. Checking the method against the conformal literature rather than against our own  
 192 earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected  
 193 in §8.

194 **Provenance discipline.** Two parser fixes were informed by inspecting rows that were already in  
 195 the test set. Rather than silently retain them, we record which fixes were test-informed and exclude  
 196 the affected family from the strict headline.

197 **Mechanical verification.** Every figure in this paper is a macro generated from a claims registry  
 198 that recomputes each value from source artifacts; no number here is hand-typed. The accompanying  
 199 repository enforces this with a test that fails if any outward-facing document contains an untraced  
 200 numeric literal.

## 10 Conclusion

Per-model prediction of quantization accuracy damage does not work on published data, and the reason is measurement noise rather than model capacity. What does work is narrower: a per-scheme calibrated envelope, validated at 90.1% coverage on unseen checkpoints, that refuses to answer where its own measured coverage is poor. The more useful contribution may be the limitation itself—the published record of quantized models is outcome-truncated, and we measured losses roughly four times worse than anything it contains. A risk estimate built on that record is a floor on risk, not a ceiling, and should say so.

## References

- Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2), 2023.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- Prashansa Gupta and Sean MacAvaney. On survivorship bias in MS MARCO. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2022. doi: 10.1145/3477495.3531832.
- Ying Jin and Emmanuel J. Candès. Confidence on the focal: Conformal prediction with selection-conditional coverage. *arXiv preprint arXiv:2403.03868*, 2024.
- Eldar Kurtić, Alexandre Marques, Mark Kurtz, and Dan Alistarh. We ran over half a million evaluations on quantized LLMs—here’s what we found. Red Hat Developer, October 2024. <https://developers.redhat.com/articles/2024/10/17/we-ran-over-half-million-evaluations-quantized-llms>.
- Yujia Tong, Yuxi Wang, Yunyang Wan, Tian Zhang, Junhao Dong, and Jingling Yuan. Does compression preserve uncertainty? a unified benchmark for quantized and sparse LLMs via conformal prediction. *arXiv preprint arXiv:2606.01850*, 2026. Wuhan University of Technology; Nanyang Technological University.
- vLLM Project. llm-compressor, 2024. <https://github.com/vllm-project/llm-compressor>.
- Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and efficient post-training quantization for large language models. *arXiv preprint arXiv:2211.10438*, 2022.
- Yuchen Zeng and Dimitris Papailiopoulos. You don’t need to run every eval. *arXiv preprint arXiv:2606.24020*, 2026. Microsoft Research, AI Frontiers.